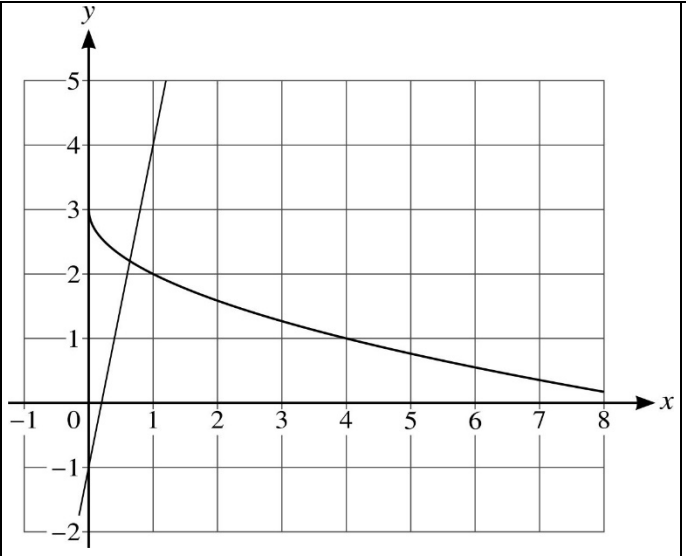
9	**(a)**			**B1** **[1]**	**1.1**	Correct line through ((0, -1) and (1, 4)	If x-intercept marked and (1, 4) slightly out may award **B1 BOD**

9	**(b)**		**DR** $3-\sqrt{x}=5x-1$ or $y=5(3-y)^2-1$ $4-5x=\sqrt{x}\Rightarrow(4-5x)^2=x$	**M1**	**3.1a**		**Or** as a quadratic in $\sqrt{x}$ $5x+\sqrt{x}-4=0$
			$25x^2-41x+16=0$ or $5y^2-31y+44=0$	**M1**	**1.1**	Getting into suitable form for solution ie =0	
			$(x-1)(25x-16)=0$ or $(5y-11)(y-4)=0$	**M1**	**1.1**	Attempt to solve quadratic by formula or factorising (oe, via $\sqrt{x}$ quadratic)	$\left(5\sqrt{x}-4\right)\left(\sqrt{x}+1\right)=0$
			$x=\frac{16}{25}=0.64$ or $y=2.2$	**A1**	**2.2a**	Correct root chosen Or replace **M1A1** with **SC1** if x = 0.64 or y = 2.2 seen with no method for solving	$\sqrt{x}=\frac{4}{5}$ so $x=0.64$
			$y=5\times0.64-1=2.2$ or $x=(3-2.2)^2=0.64$	**M1**	**1.1**	FT *their* positive root	
			$\frac{1}{2}(0.64-0.2)(2.2)$ $\left[=0.484=\frac{121}{250}\right]$	**M1**	**3.1a**	Attempt to find area of triangle	$\frac{1}{2}\times0.64\times2.2$ is M0
			Alternative method for area of triangle $\int_{0.2}^{0.64}(5x-1)\,dx$	**M1** **M1**		Correct integral Correct limits (FT their positive root)	
			$\int_{0.64}^{4}\left(3-\sqrt{x}\right)\,dx$	**M1***	**2.1**	Allow any limits $0\le x\le4$	Allow if clearly embedded eg $\int_{0.64}^{4}\left(5x-4+\sqrt{x}\right)\,dx$
			$\left[3x-\frac{2}{3}x^{\frac{3}{2}}\right]_{0.64}^{4}$	**M1****	**1.1**	Integration of M1* integral $\sqrt{x}$ term correct (ignore limits)	Dep on M1* only
			$\left(12-\frac{16}{3}\right)-\left(1.92-\frac{128}{375}\right)=\frac{636}{125}=5.088$	**M1**	**1.1**	Evaluation of M1* integral substitution seen	Dep on M1**
			Total area = 0.484 + 5.088 = 5.572 = $\frac{1393}{250}=5\frac{143}{250}$	**A1**	**1.1**	All correct, other partitions possible	Dep on all 9 previous marks
				[10]			